

ABM BUSINESS LOGIC BIBLE

SECTION 4 — INTEGRATIONS & EXTERNAL INTERFACES

CHECKPOINT 4.1 — INT-SIG · ACCOUNT INTELLIGENCE ACQUISITION

ENGINE

The deepest integration domain. INT-SIG is not a feed reader — it is the engine that continuously produces, for every target account, a **living Master Intelligence Dossier** and resolves the **connection path back to Decimal**. It automates the manual BD-intelligence methodology already proven on SNB, Riyadh Bank and Al Rajhi Bank.

Sub-streams	8	5-Whys chains	8 (root-caused)
New rules (INT-SIG)	34	Headline stream	SIG-PATH → Decimal

WHY THIS DOMAIN LEADS SECTION 4

A signal with no warm path is an **unactionable signal**. The entire engine exists to convert intelligence into a reachable conversation — so the connection-to-Decimal path (SIG-PATH) is the acquisition *target*, not a byproduct. Every other sub-stream feeds it.

INT-SIG must hit the same bar as the manual dossier: **named** contacts with verification badges, vendor-gap analysis, subsidiary activity, cross-group power-node detection, and per-account exclusivity discipline (Backbase = SNB only; Efidence = Riyadh only). A weaker automated bar is a regression.

4.1.0 INT-SIG ARCHITECTURE — EIGHT INTELLIGENCE SUB-STREAMS

Each sub-stream is a specified harvester that runs per target account on its own cadence, normalizes into a single account intelligence graph, and emits both scoring triggers and dossier updates. The Riyadh Bank column shows the concrete acquisition target for each stream.

Sub-stream	Acquires (Riyad Bank example)	Source layer	Cadence
SIG-NEWS	Every news item, announcement, press release, internal circular surfaced publicly	Google News API · SAMA RSS · bank newsroom scrape	Hourly
SIG-EVENT	Events, seminars, conferences, awards, sponsorships, speaking slots	LinkedIn org page · event sites · awards bodies	Daily
SIG-FIN	Financial position, quarterly results, full annual report, credit ratings	Tadawul disclosures · bank IR page · rating agencies	On-publish + quarterly
SIG-REG	Regulatory circulars, SAMA directives, licences, compliance moves	SAMA · CMA · Tadawul	Daily
SIG-EXEC	Full C-suite dossier: role, tenure, bio, public speeches/quotes, moves	LinkedIn · conference agendas · interviews · news	Weekly + event-driven
SIG-VENDOR	Vendor relationships + their activity (e.g. Efigence at Riyadh)	Annual report · press · case studies · job posts	Monthly + event-driven
SIG-SUBS	Subsidiaries and their activity	Annual report · subsidiary sites · news	Monthly + event-driven
SIG-PATH	Connection paths to Decimal: LinkedIn degree, mutual connections, shared alumni/vendor/partner nodes, email pattern	LinkedIn graph · enrichment · Decimal employee-network ingestion	Continuous

OUTPUT CONTRACT — THE LIVING DOSSIER

INT-SIG's output is a versioned **Account_Intelligence_Dossier** object per account, structurally identical to the manual Master Intelligence Dossier: overview · C-suite (named, badged) · subsidiaries (named execs) · vendor registry + gap analysis · connectors/warm paths · Named Individuals Index. Every downstream module (scoring, enrichment, message generation, handoff brief) reads from this single object.

4.1.1 SIG-PATH — CONNECTION-TO-DECIMAL RESOLUTION (headline sub-stream)

SIG-PATH is resolved against the **full set**: Decimal's own employee LinkedIn/email network (ingested as a graph) *and* the generalized connector patterns already identified across Saudi banks (Mastercard/KPMG/Deloitte alumni bridges, Tarabut, Fintech Saudi/SAMA initiatives, and the IT-subsidary gatekeeper pattern).

SIG-PATH — Connection-to-Decimal Resolution

Why #1: Why must SIG-PATH be a first-class stream, not a lookup at send time?

Because the warm path determines whether a signal is actionable at all, and which message/sender to use. Resolving it lazily at send time means the engine scores and prioritizes accounts it may not be able to reach — wasting attention capital on unreachable targets.

Why #2: Why resolve against Decimal's own network and not just generalized connectors?

Because a 1st-degree LinkedIn edge from a Decimal employee to a target executive is the strongest possible path and is invisible to generalized patterns. Ignoring Decimal's own graph discards the highest-confidence warm paths the business already owns.

Why #3: Why must paths be re-resolved continuously rather than cached?

Because paths decay: people change jobs (the Market Memory / career-transition layer), connections are added, connectors move firms. A path discovered yesterday may be broken today, and a new path may have just opened — which is itself a signal.

Why #4: Why treat a broken executive path as a signal rather than silent data loss?

Because an executive leaving a target both breaks an existing warm path and creates two events: a vacancy at the target (intent signal) and a relationship to repair or follow the person to their new employer.

Why #5: Why generalize the cross-group power node and IT-gatekeeper patterns into rules?

Because one person sitting across multiple target entities (the Hossam Al Basrawi pattern) is the single most efficient relationship, and routing through the IT-subsidary gatekeeper (EJADA-equivalent) before the C-suite is the proven sequence. Hard-coding these turns proven tradecraft into deterministic engine behaviour.

Root cause: SIG-PATH must continuously maintain a ranked, decaying set of warm paths per target — sourced from Decimal's own graph plus generalized connectors — and emit path-open / path-break events as first-class signals.

Priority	Path type	Resolution source	Confidence basis
P1	LinkedIn 1st-degree (Decimal employee ↔ target)	Decimal network ingestion	Direct edge — highest
P2	Mutual connection (2nd-degree via Decimal person)	Decimal network + enrichment	Shared-node count + node seniority
P3	Shared node: ex-employer / alumni / vendor / board	Enrichment + SIG-VENDOR/SUBS	Strength of shared affiliation
P4	Generalized connector bridge (KPMG/Mastercard/Tarabut)	Connector pattern registry	Connector's confirmed reach
P5	Cold email pattern (no warm path)	SIG-EXEC email pattern	Deliverability only — lowest

New rules

Rule ID	Specification
SIG-PATH-01	Path priority is strictly ordered P1>P2>P3>P4>P5 (LinkedIn 1st-degree > mutual connection > shared node > generalized connector > cold email). Highest available path is selected for sender/route assignment.
SIG-PATH-02	Decimal employee-network ingestion: the engine maintains an opt-in graph of Decimal personnel LinkedIn connections + email domains, refreshed continuously, used as the P1/P2 resolution source. Personal network data is access-controlled and never exported off-platform.
SIG-PATH-03	Every warm path carries a decay TTL. On expiry the path is re-resolved; an unresolvable previously-valid path emits a path-break event.
SIG-PATH-04	Executive departure from a target emits BOTH a target-vacancy intent signal (to SIG-EXEC/scoring) AND a follow-the-person event (re-anchor the relationship at the new employer).
SIG-PATH-05	Cross-group power-node detection: any individual resolved as holding roles across ≥ 2 target/priority entities is flagged as a high-value node and surfaced to every account they touch.
SIG-PATH-06	IT-subsidiary gatekeeper rule: where a target has a wholly-/majority-owned implementation arm (EJADA-equivalent), no C-suite outreach is sequenced until the gatekeeper relationship is established or explicitly waived by a human.
SIG-PATH-07	Mutual-connection confidence = $f(\text{shared-node count, shared-node seniority, recency of connection})$. Paths below a confidence floor are demoted to P4/P5.
SIG-PATH-08	Generalized connector registry (KPMG/Mastercard/Deloitte/Tarabut/Fintech-Saudi) is maintained as reusable P4 bridges with confirmed-reach metadata; a connector's reach is per-bank-verified, never assumed global.
SIG-PATH-09	No warm path above P5 $\rightarrow$ account is flagged 'reachable: cold only'; scoring applies an unreachability penalty so attention is not over-allocated to unreachable targets.
SIG-PATH-10	SIG-PATH never fabricates a path or a mutual connection. Unverifiable paths are recorded as ~inferred and excluded from automated sender assignment until human-confirmed.

4.1.2 SIG-EXEC — C-SUITE DOSSIER & SPEECH CAPTURE

SIG-EXEC — C-Suite Dossier & Speech Capture

Why #1: Why capture full executive dossiers including public speeches and quotes?

Because message personalization and strategic relevance depend on what an executive has publicly said they care about. A quote from a CDO's conference keynote is the strongest possible personalization hook and proves the outreach is researched, not templated.

Why #2: Why resolve to named persons with verification badges rather than role titles?

Because 'the CTO' is not reachable — a named person with a confirmed LinkedIn and email pattern is. Role-level desks are a fallback only, and conflating two sources that disagree on who holds a role produces wrong-person outreach that burns the account.

Why #3: Why weekly + event-driven refresh rather than one-time enrichment?

Because executive moves are frequent and high-impact: a new CTO is a vendor-evaluation window (P1 signal) and simultaneously invalidates the prior dossier and any path anchored on the departed exec.

Why #4: Why must conflicting role-holders be flagged, never asserted?

Because asserting a contested name confidently (the Emkan CEO ambiguity) causes the engine to address the wrong person with false confidence — worse than admitting uncertainty.

Why #5: Why tie SIG-EXEC output directly to the message-generation module's grounding set?

Because the hallucination-prevention rules require every personalization claim to be grounded in a captured, sourced fact; SIG-EXEC is the authoritative supply of those grounded executive facts.

Root cause: SIG-EXEC must produce named, badged, sourced executive dossiers with captured public speech/quote material, refreshed on executive-move events, feeding both SIG-PATH (anchors) and message grounding.

New rules

Rule ID	Specification
SIG-EXEC-01	Every C-suite/decision-maker contact resolves to a real named person where publicly findable, with a verification badge (✓✓ 2+ sources, ✓ 1 source, ~ inferred, ■ sources conflict, ✗ not found). Role-level desk only as labelled fallback with LinkedIn search string + email pattern.
SIG-EXEC-02	Public speeches, keynote quotes, interview statements are captured with source + date and stored as grounded personalization material linked to the named exec.
SIG-EXEC-03	Conflicting role-holders are flagged ■ verify and excluded from automated outreach until resolved; the engine never asserts one contested name confidently.
SIG-EXEC-04	Executive-move detection triggers: (a) dossier re-resolution, (b) target-vacancy intent signal, (c) SIG-PATH re-anchor / follow-the-person.
SIG-EXEC-05	Only SIG-EXEC-sourced, badged facts may be used as grounding for message personalization; ungrounded executive claims are rejected by QC.

4.1.3 SIG-VENDOR & SIG-SUBS — VENDOR REGISTRY, GAP ANALYSIS & SUBSIDIARY ACTIVITY

SIG-VENDOR — Vendor Registry & Displacement-Gap Analysis

Why #1: Why track the bank's vendors and their activity, not just the bank itself?

Because Decimal's opportunity is defined by the incumbent stack: knowing Riyadh runs Efigence (and SNB runs Backbase) is what locates the displacement or complement gap that the pitch targets.

Why #2: Why per-account exclusivity discipline on vendor facts?

Because vendor relationships are bank-specific. Asserting an SNB-only vendor (Backbase) at Riyadh, or reusing a 12-year ITC Infotech relationship across banks, produces a factually wrong pitch that destroys credibility.

Why #3: Why is a vendor's own activity (hiring, case studies, contract news) a signal?

Because incumbent expansion or distress at the vendor changes the displacement window — a struggling or over-extended incumbent is an opening; a freshly renewed one closes it.

Why #4: Why feed vendor-gap analysis into scoring rather than only into messaging?

Because the size and freshness of the displacement gap is a core attractiveness driver, not just a talking point — it belongs in Effective Opportunity, not only in the message.

Why #5: Why route vendor intelligence through the same grounded-fact contract as SIG-EXEC?

Because a fabricated or stale vendor claim is as damaging as a wrong executive name; both must be sourced, badged, and account-scoped.

Root cause: SIG-VENDOR must maintain a per-account, exclusivity-disciplined vendor registry with displacement-gap analysis that feeds both scoring and grounded messaging; SIG-SUBS does the same for subsidiaries and their activity.

New rules

Rule ID	Specification
SIG-VENDOR-01	Vendor registry is strictly per-account; a vendor fact confirmed at one bank is never propagated to another without independent confirmation (exclusivity discipline).
SIG-VENDOR-02	Each vendor entry carries a Decimal displacement/complement gap classification (displace · complement · no-fit) with source + freshness.
SIG-VENDOR-03	Vendor activity events (incumbent hiring surge, contract renewal/loss, case study) update the displacement-window state and may emit scoring triggers.
SIG-VENDOR-04	Displacement-gap magnitude + freshness feeds the account's Effective Opportunity, not only message content.
SIG-SUBS-01	Subsidiaries are resolved from the bank's official annual-report PDF (authoritative); subsidiary execs resolved to named persons (Phase-7 discipline) with badges.
SIG-SUBS-02	Subsidiary activity (launches, hires, regulatory moves) is ingested as account-linked signals; an IT/implementation subsidiary is additionally tagged as a SIG-PATH gatekeeper node.

4.1.4 SIG-NEWS · SIG-EVENT · SIG-FIN · SIG-REG — DISCLOSURE & ACTIVITY STREAMS

These four streams share a common 5-Whys spine: *why ingest* → each is a dated, sourced intent or context signal; *why normalize to one graph* → the same event surfaces across multiple sources and must deduplicate to one account-linked fact; *why freshness TTL* → a stale disclosure mis-scores an account; *why source-trust weighting* → a SAMA circular and a forum rumour cannot carry equal weight; *root* → all four must resolve, dedup, badge, and time-decay into the single account intelligence graph that scoring reads.

Stream	Primary signal meaning	Scoring trigger example
SIG-NEWS	Announcements, circulars, press — context + intent	New digital initiative announced → priority bump
SIG-EVENT	Events/awards/speaking — access + timing	Exec speaking at fintech summit → SIG-EXEC quote + event window
SIG-FIN	Financial position — budget availability	Strong quarter / new annual report → infrastructure budget signal
SIG-REG	SAMA/CMA circulars — compliance urgency	Open-banking directive → compliance-urgency window opens

New rules

Rule ID	Specification
SIG-NEWS-01	All four streams normalize + entity-resolve + deduplicate into one account intelligence graph; the same event from N sources collapses to one badged fact (badge rises with corroborating sources).
SIG-NEWS-02	Every ingested item carries source + date + a freshness TTL; expired items are demoted from active scoring but retained in Market Memory.
SIG-REG-01	Source-trust weighting: regulator (SAMA/CMA/Tadawul) > bank official > tier-1 press > aggregator > forum/rumour. Low-trust-only facts cannot cross the scoring-trigger threshold alone.
SIG-FIN-01	Financial disclosures (quarterly results, annual report, rating changes) map to budget-availability signals consumed by Effective Opportunity's Budget Modifier.
SIG-REG-02	Regulatory circulars map to compliance-urgency windows with explicit open/close timing fed to the Timing module (Window Multiplier).
SIG-EVENT-01	Event/speaking detections cross-link to SIG-EXEC (capture the speech as grounded material) and to the Timing module (event proximity = engagement window).

4.1.5 INT-SIG CONSOLIDATED RULE INDEX (34 new rules)

All rules surfaced in this checkpoint, grouped by sub-stream. These are additive to the Section 1–3 rule base and will carry forward into the unified Section 4 document alongside the remaining ten integration domains (INT-ENR, INT-LIN, INT-EML, INT-CRM, INT-SLK, INT-ANL, INT-ORC, API, FAIL, OBS).

Group	IDs	Count	Theme
SIG-PATH	01–10	10	Connection-to-Decimal: priority ladder, Decimal-network ingestion, decay, power-node + gatekeeper, anti-fabrication
SIG-EXEC	01–05	5	Named/badged dossiers, speech capture, move detection, grounded-fact contract
SIG-VENDOR	01–04	4	Per-account registry, displacement gap, activity events, scoring feed
SIG-SUBS	01–02	2	Annual-report-authoritative subsidiaries, activity + gatekeeper tagging
SIG-NEWS	01–02	2	Dedup to one graph, freshness TTL
SIG-REG	01–02	2	Source-trust weighting, compliance-window mapping
SIG-FIN	01	1	Disclosure → budget-availability signal
SIG-EVENT	01	1	Event ↔ SIG-EXEC + timing window cross-link

NEXT — UNIFIED SECTION 4 BUILD

With INT-SIG approved, the unified Section 4 PDF will incorporate this domain in full plus the remaining ten: **INT-ENR** (enrichment fallback chain), **INT-LIN** (Heyreach platform-policy limits + ban-risk circuit breaker), **INT-EML** (SPF/DKIM/DMARC + warmup gating), **INT-CRM** (idempotent HubSpot sync + authority split), **INT-SLK**, **INT-ANL**, **INT-ORC** (no-business-logic n8n boundary), **API** (5 service contracts), **FAIL** (retry/DLQ/circuit-breaker matrix), and **OBS** (observability + PDPL/compliance) — each via full 5-Whys, totalling ~77 additional rules.